1. We have

$$\lim_{h\to 0}\frac{\frac{x+h}{x+h-1}-\frac{x}{x-1}}{h}=\lim_{h\to 0}\frac{\frac{x^2-x+hx-h-x^2-hx+x}{(x+h-1)(x-1)}}{h}$$
$$=\lim_{h\to 0}\frac{-1}{(x+h-1)(x-1)}$$
$$=\frac{-1}{(x-1)^2}.$$

The limit exists for all x except $x=1$.

2. If $f:\mathbb{R}^n\to\mathbb{R}$ is a differentiable function, the function $\nabla f(x):\mathbb{R}^n\to\mathbb{R}^n$ is called the gradient of f and is defined as

$$\nabla f(x_1,x_2,x_3)=\begin{pmatrix}\frac{\partial f}{\partial x_1}\\ \frac{\partial f}{\partial x_2}\\ \frac{\partial f}{\partial x_3}\end{pmatrix}.$$

Therefore, the gradient of our function is

$$\nabla f(x_1,x_2,x_3)=\begin{pmatrix}e^{x_1}+2x_1x_3-x_2x_3\\ -x_1x_3\\ x_1^2-x_1x_2\end{pmatrix}.$$

3. Let $f:\mathbb{R}^n\to\mathbb{R}$ be a differentiable function. Consider $x\in\mathbb{R}^n$ and $d\in\mathbb{R}^n$. The directional derivative of f in x in the direction d is given by

$$\lim_{\alpha\to 0}\frac{f(x+\alpha d)-f(x)}{\alpha},$$

if the limit exists. In addition, when the gradient exists, the directional derivative is the scalar product between the gradient of f and the direction d, that is,

$$\nabla f(x)^T d.$$

Therefore, the directional derivative is

$$(d_1\ d_2\ d_3)\nabla f(x_1,x_2,x_3)=$$

$$d_1(e^{x_1}+2x_1x_3-x_2x_3)-d_2x_1x_3-d_3(x_1^2-x_1x_2).$$

4. This is incorrect. Indeed, the gradient is

$$\nabla f(x_1,x_2)=\begin{pmatrix}2x_1\\ 2x_2\end{pmatrix}$$

and the point $(0,0)$, we obtain as directional derivative:

$$\nabla f(0,0)^T d=(0,0)\begin{pmatrix}1\\ 1\end{pmatrix}=0.$$

5. Recall that the direction opposite the gradient is that in which the function has its steepest descent.

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} x_1 \\ 4x_2 \end{pmatrix}.$$

In point $x = (1\,1)^T$

$$\nabla f(x) = \begin{pmatrix} x_1 \\ 4x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix},$$

The directional derivative in f in the direction opposite to the gradient is:

$$-\nabla f(x)^T \nabla f(x) = \begin{pmatrix} -1 & -4 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} = -17$$

6. First, we are going to calculate the gradient matrix:

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} \\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 2x_1x_2 & -\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} \\ x_1^2 & -2x_2\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} \end{pmatrix}.$$

Knowing that transposing the gradient we obtain the Jacobian, we are going to calculate it

$$JF(x_1, x_2) = \nabla f(x)^T = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 2x_1x_2 & x_1^2 \\ -\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} & -2x_2\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} \end{pmatrix}.$$

7. The function $f(x, y) = 400 - 3x^2y^2$ is differentiable in point $(1, -2)$, therefore the gradient is going to give us the direction in which the depth has its steepest ascent.

$$\nabla f(x, y) = (-6xy^2, -6x^2y)$$

$$\nabla f(1, -2) = (-24, 12).$$

The normalized direction in which the depth increases as quick as possible is then $(-2/\sqrt{5}, 1/\sqrt{5})$.